

The empirical recurrence for OEIS A233854: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A233854 records “Empirical recurrence of order 68”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A233854 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having the sum of the squares of all six edge and diagonal differences equal to 59.”. It has offset 1 and begins

1040, 4824, 22472, 129728, 699984, 4332592, 24793848, 156146264, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:01:38, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	7	c_{18}	−9121616340914	c_{35}	−785232928547041790	c_{52}	524259480899595520
c_2	168	c_{19}	−77957551893466	c_{36}	4087911230555934282	c_{53}	−198408039904250112
c_3	−1270	c_{20}	99919755139317	c_{37}	787021014621045446	c_{54}	−149033418336400384
c_4	−12288	c_{21}	521576445413875	c_{38}	−6339582058210470200	c_{55}	61491552597029888
c_5	103188	c_{22}	−811056852001059	c_{39}	−378801025839773604	c_{56}	31851595992698880
c_6	514208	c_{23}	−2820921354407267	c_{40}	8131733259053299596	c_{57}	−14360343594446848
c_7	−5006592	c_{24}	5122316781773365	c_{41}	−377744599002526420	c_{58}	−4882874074316800
c_8	−13461030	c_{25}	12384883345574551	c_{42}	−8608472651916260352	c_{59}	2455295231311872
c_9	163214468	c_{26}	−25810019428225854	c_{43}	1135689966229914232	c_{60}	494102321627136
c_{10}	219474480	c_{27}	−44196562049965212	c_{44}	7493503961049598592	c_{61}	−293573589270528
c_{11}	−3813960552	c_{28}	105309729739876873	c_{45}	−1506877320716023808	c_{62}	−27151792078848
c_{12}	−1785958324	c_{29}	127935053683121497	c_{46}	−5334267194675177936	c_{63}	22776467423232
c_{13}	66547193722	c_{30}	−351239863423144532	c_{47}	1373330575619557552	c_{64}	195513286656
c_{14}	−9872832979	c_{31}	−298365384907904966	c_{48}	3081316074909520480	c_{65}	−1003191533568
c_{15}	−891492039593	c_{32}	963485722667903092	c_{49}	−936648587399384000	c_{66}	53678702592
c_{16}	543645257182	c_{33}	552286454466980220	c_{50}	−1428991878743800704	c_{67}	18572378112
c_{17}	9352966688310	c_{34}	−2181960884231780886	c_{51}	490680556002479424	c_{68}	−1811939328

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68 + 4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

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